

Somanshu Mohapatra

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SKILLS

Languages:	C++, Python, Java, JavaScript, HTML, SQL, Embedded C, MATLAB, GNU Octave.
AI & Vision:	AIML, Deep Learning (PyTorch, TensorFlow, CNN, YOLO), OpenCV, OpenVINO, AI Image Generation (Stable Diffusion, AUTOMATIC1111, Illustrious Models).
Robotics:	Forward/Inverse Kinematics (DH Parameters), ROS2, RViz2, MoveIt, Gazebo, Path Planning, Motion Control, PID Control, Sensor Fusion, SLAM.
Electronics:	Arduino UNO, Raspberry Pi, STM32, ATmega328P, RP2040, Motor Drivers, Serial Communication (UART, I2C, SPI), LabVIEW, Proteus, NGspice, Pspice.
CAD & Design:	SolidWorks, Fusion 360, AutoCAD, Creo Parametric, ANSYS HFSS, Assembly Design, Rapid Prototyping, 3D Printing (FDM/SLA), Bambu Slicer, LightBurn.
Soft Skills:	Effective Organizer, Team Management, Strong Interpersonal Skills, Problem Solving.

TRAININGS

AutomateX: Mastering PLCs for Industrial Automation LPU	Mar' 25
- Acquired comprehensive knowledge of PLC architecture, ladder logic programming, and industrial automation principles. - Simulated automated industrial workflows using TwidoSuite to manage sensor inputs and heavy actuator outputs. - Tested fail-safe mechanisms and sequence patterns to reinforce practical understanding of factory floor operations.	

PROJECTS

2.4GHz Microstrip Patch Antenna ANSYS HFSS	Oct' 25
- Designed and simulated a high-frequency microstrip patch antenna optimized for 2.4GHz wireless communication applications. - Conducted electromagnetic field analysis using ANSYS HFSS to evaluate return loss, gain, and radiation patterns. - Optimized physical dimensions and substrate selection to achieve superior impedance matching and antenna efficiency.	
Dynamic Audio Monitoring and Narration Raspberry Pi, Python, AssemblyAI API	May' 25
- Developed a real-time transcription system for the deaf using Raspberry Pi to provide accessible communication. - Integrated AssemblyAI API and WebSockets with an I2C SSD1306 OLED display utilizing multi-threading technology. - Optimized inference speeds on Raspberry Pi 4B by deploying an Intel AI Accelerator, significantly reducing latency.	
3-DOF Robotic Arm with Real-Time Control MATLAB, SolidWorks, Arduino IDE, Bambu Studio	Apr' 24
- Constructed a 3-DOF robotic arm utilizing 3D-printed parts to optimize the payload-to-weight ratio. - Programmed real-time motor control via MATLAB-UART communication, enabling precise joint actuation. - Targeted future implementation of Forward and Inverse Kinematics to automate complex trajectory planning.	
Python-SQL Accounting Software Python, MySQL, CLI	Jun' 23
- Engineered a database-driven accounting CLI application to manage financial records and client transactions. - Implemented CRUD operations using Python-MySQL connectivity to facilitate seamless data management. - Provided a robust interface for searching records by client name or date, streamlining data retrieval.	

CERTIFICATES

Data Base Management System NPTEL	Aug' 25
Fusioncraft Resolute Lab	Mar' 24

ACHIEVEMENTS

Patent: Method for Enhancing Food Accessibility and Energy Efficiency in Refrigeration	Sep' 25
Application No.: 202511085342 A	

EDUCATION

Lovely Professional University	Phagwara, Punjab
BTech Robotics and Automation; CGPA: 8.38	Aug' 23 – Present
Shri S.N. Siddeshwar Public School	Gurugram, Haryana
Intermediate, PCMB; Percentage: 85.6%	Mar' 21 – May' 23
Ryan International School	Gurugram, Haryana
Matriculation; Percentage: 81.6%	Mar' 20 – May' 21